

SHOALWATER, Australia

WMO: 953700

Lat: 22.470S

Lon: 150.179E

Elev: 28

StdP: 100.99

Time Zone: 10.00 (E10)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.8	6.4	-1.7	3.3	15.7	0.8	4.0	13.4	7.3	17.1	6.7	17.5	0.5	160	0.326

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.2	32.4	23.7	31.6	23.5	30.9	23.2	26.3	29.0	25.7	28.5	25.1	28.2	4.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.7	21.1	27.4	24.9	20.0	27.0	24.2	19.2	26.8	82.5	28.8	79.4	28.8	77.1	28.3	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	6.1	5.5	2.0	34.8	1.5	1.4	0.9	35.8	0.0	36.6	-0.8	37.4	-1.9	38.4		
			1.2	27.2	2.0	1.5	-0.2	28.3	-1.3	29.1	-2.4	30.0	-3.9	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.2	26.9	26.5	25.7	23.2	19.8	17.3	16.6	17.3	20.1	22.6	24.8	26.4
	DBStd	4.24	1.34	1.34	1.52	1.64	2.34	2.62	2.70	2.50	2.15	1.98	1.69	1.47
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	187	0	0	0	1	14	49	67	49	7	1	0	0
	CDD10.0	4467	523	463	486	396	303	220	204	227	302	390	444	509
	CDD18.3	1611	265	229	227	146	58	18	13	17	59	132	194	251
	CDH23.3	13352	2611	2152	1923	928	316	67	63	139	422	868	1514	2348
CDH26.7	3582	853	665	530	157	23	1	1	5	44	142	400	761	
Wind	WSAvg	2.6	3.0	2.9	2.8	2.5	2.2	2.2	2.1	2.2	2.4	2.8	2.9	3.0
Precipitation	PrecAvg	823	155	148	91	61	38	39	25	27	20	35	57	125
	PrecMax	1901	400	463	360	227	146	154	155	162	124	98	285	528
	PrecMin	374	30	15	1	0	0	1	0	0	0	0	0	0
	PrecStd	295	100	123	93	64	34	33	39	38	29	27	63	119
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	33.2	32.6	30.3	28.5	26.3	26.0	27.2	29.7	30.5	32.3	33.4
		MCWB	24.5	24.5	23.9	22.2	20.4	18.9	18.1	18.3	19.5	21.1	22.1	22.3
	2%	DB	32.1	32.2	31.2	29.2	27.2	25.0	24.9	25.8	27.7	29.0	30.6	31.8
		MCWB	24.0	24.1	23.7	21.8	19.7	18.2	17.6	17.9	19.4	21.0	22.1	22.9
	5%	DB	31.2	31.2	30.3	28.4	26.3	24.0	24.0	24.9	26.8	28.2	29.7	31.0
		MCWB	23.7	23.9	23.4	21.5	19.4	17.8	17.3	17.5	18.9	20.5	21.7	22.7
	10%	DB	30.4	30.2	29.5	27.6	25.2	23.0	22.9	23.8	25.8	27.4	29.0	30.2
		MCWB	23.4	23.7	23.0	21.3	19.0	17.5	16.9	17.0	18.6	20.0	21.5	22.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	27.6	26.5	26.2	23.5	21.2	21.5	21.2	22.6	23.5	25.9	25.9
		MCDB	29.9	29.7	29.0	27.8	25.6	23.6	22.7	24.4	26.4	27.3	28.1	29.7
	2%	WB	26.2	26.5	25.6	24.9	21.9	20.2	20.2	20.4	21.9	22.8	24.5	25.2
		MCDB	28.9	29.0	28.3	26.9	24.5	22.7	23.0	23.4	24.8	26.2	27.5	28.7
	5%	WB	25.5	25.5	25.0	23.8	21.0	19.5	19.3	19.4	21.1	22.2	23.4	24.7
		MCDB	28.5	28.6	27.9	26.0	23.9	22.1	22.0	22.9	24.4	26.0	27.1	28.1
	10%	WB	24.8	24.9	24.3	22.7	20.2	18.6	18.3	18.4	20.2	21.5	22.8	24.2
		MCDB	28.2	28.1	27.5	25.5	23.5	21.7	21.1	22.3	23.8	25.5	26.9	27.7
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.1	7.7	9.1	11.1	11.5	12.9	13.6	12.0	10.0	8.5	7.6
		MCDBR	8.5	8.9	8.9	10.2	12.1	12.5	13.4	14.0	12.8	10.7	9.4	8.5
		MCWBR	2.4	2.7	2.8	4.3	5.4	6.4	7.0	7.0	5.5	4.0	3.0	2.4
	5% WB	MCDBR	7.0	7.0	7.4	8.1	9.1	9.1	9.2	10.8	9.4	8.7	7.9	6.9
		MCWBR	2.7	2.9	3.0	4.4	4.9	5.0	5.4	6.0	4.1	3.6	2.7	2.7
Clear-Sky Solar Irradiance	taub	0.398	0.390	0.374	0.352	0.331	0.321	0.311	0.324	0.351	0.377	0.391	0.394	
	taud	2.510	2.538	2.570	2.610	2.623	2.647	2.651	2.596	2.513	2.472	2.467	2.492	
	Ebn at Noon	947	941	931	911	893	884	906	927	938	943	947	951	
	Edh at Noon	115	109	101	90	82	77	79	90	106	115	119	117	
All-Sky Solar Radiation	RadAvg	6.44	5.99	5.48	4.75	4.10	3.59	4.07	4.93	5.99	6.62	6.84	6.81	
	RadStd	0.43	0.64	0.46	0.43	0.23	0.33	0.30	0.33	0.41	0.40	0.73	0.68	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page